

```
lab task 1 lab7.cpp
1 #include<iostream>
2 #include<string>
3 using namespace std;
4 class robot
5 {
6     public:
7         string robotID;
8         string robotType;
9         int battery;
10        float speed;
11    };
12    int main()
13    {
14        robot r1, r2, r3;
15        r1.robotID="R-001";
16        r1.robotType="picker";
17        r1.battery=90;
18        r1.speed=1.5;
19        r2.robotID="R-002";
20        r2.robotType="scout";
21        r2.battery=55;
22        r2.speed=2.2;
23        r3.robotID="R-003";
24        r3.robotType="carrier";
25        r3.battery=20;
26        r3.speed=0.8;
27        cout << "-----ROBOT FLEET STATUS-----"<<endl;
28        cout << "ID:" << r1.robotID
29              << " TYPE:" << r1.robotType
30              << " BATTERY:" << r1.battery<< " %"
31              << " SPEED:" << r1.speed<< "m/s"<< endl;
32        cout << "ID:" << r2.robotID
33              << " TYPE:" << r2.robotType
34              << " BATTERY:" << r2.battery<< " %"
35              << " SPEED:" << r2.speed<< "m/s"<< endl;
36        cout << "ID:" << r3.robotID
37              << " TYPE:" << r3.robotType
38              << " BATTERY:" << r3.battery<< " %"
39              << " SPEED:" << r3.speed<< "m/s"<< endl;
40        return 0;
41    }
```

```
C:\Users\Taha's Lappy\OneDrive\Documents\oop lab 7\lab task 1 lab7.exe
-----ROBOT FLEET STATUS-----
ID:R-001| TYPE:picker| BATTERY:90%| SPEED:1.5m/s
ID:R-002| TYPE:scout| BATTERY:55%| SPEED:2.2m/s
ID:R-003| TYPE:carrier| BATTERY:20%| SPEED:0.8m/s

-----
Process exited after 0.1851 seconds with return value 0
Press any key to continue . . .
```

```
lab task 1 lab7.cpp labtask2 lab 7.cpp
1 #include<iostream>
2 #include<string>
3 using namespace std;
4 class sensor
5 {
6     public:
7         string sensorID;
8         string sensorType;
9         float reading;
10        bool isactive;
11    };
12    int main()
13    {
14        sensor s1,s2,s3;
15        s1.sensorID="sensor-001";
16        s1.sensorType="ultrasonic";
17        s1.reading=25.4;
18        s1.isactive="true";
19        s2.sensorID="sensor-002";
20        s2.sensorType="infrared";
21        s2.reading=10.01;
22        s2.isactive="false";
23        s3.sensorID="sensor-003";
24        s3.sensorType="camera";
25        s3.reading=0;
26        s3.isactive="true";
27        cout << "-----sensors status report-----"<<endl;
28        cout << boolalpha;
29        cout << "ID:" << s1.sensorID
30              << " Type:" << s1.sensorType
31              << " Reading:" << s1.reading
32              << " Active:" << s1.isactive << endl;
33        cout << "ID:" << s2.sensorID
34              << " Type:" << s2.sensorType
35              << " Reading:" << s2.reading
36              << " Active:" << s2.isactive << endl;
37        cout << "ID:" << s3.sensorID
38              << " Type:" << s3.sensorType
39              << " Reading:" << s3.reading
40              << " Active:" << s3.isactive << endl;
41        return 0;
42    }
43
```

```
C:\Users\Taha's Lappy\OneDrive\Documents\oop lab 7\labtask2 lab 7.exe
-----sensors status report-----
ID: sensor-001 | Type: ultrasonic | Reading: 25.4 | Active: true
ID: sensor-002 | Type: infrared | Reading: 10.01 | Active: true
ID: sensor-003 | Type: camera | Reading: 0 | Active: true

-----
Process exited after 0.1552 seconds with return value 0
Press any key to continue . . .
```

```

4 class AModel
5 {
6     public:
7         string modelName;
8         string version;
9         string algorithm;
10        float accuracy;
11        bool isDeployed;
12    };
13    int main()
14    {
15        AModel m1,m2,m3;
16        m1.modelName="ResNet-50";
17        m1.version="v2.1";
18        m1.algorithm="CNN";
19        m1.accuracy=0.95;
20        m1.isDeployed=true;
21        m2.modelName="YOLOv8";
22        m2.version="v8.0";
23        m2.algorithm="CNN";
24        m2.accuracy=0.97;
25        m2.isDeployed=true;
26        m3.modelName="SpamDetector";
27        m3.version="v9.0";
28        m3.algorithm="SVM";
29        m3.accuracy=0.89;
30        m3.isDeployed=false;
31        cout <<boolalpha;
32        cout <<"-----AI MODEL CARD-----"<< endl;
33        cout << "NAME : "<<m1.modelName<<endl;
34        cout << "VERSION : "<<m1.version<<endl;
35        cout << "ALGORITHM : "<<m1.algorithm<<endl;
36        cout << "ACCURACY : "<<m1.accuracy<<endl;
37        cout << "DEPLOYED : "<<m1.isDeployed<<endl;
38        cout <<"-----AI MODEL CARD-----"<< endl;
39        cout << "NAME : "<<m2.modelName <<endl;
40        cout << "VERSION : "<<m2.version<<endl;
41        cout << "ALGORITHM : "<<m2.algorithm <<endl;
42        cout << "ACCURACY : "<<m2.accuracy << endl;
43        cout << "DEPLOYED : "<<m2.isDeployed <<endl;
44        cout <<"-----AI MODEL CARD-----"<< endl;
45        cout << "NAME : "<<m3.modelName<<endl;
46        cout << "VERSION : "<<m3.version<<endl;
47        cout << "ALGORITHM : "<<m3.algorithm<<endl;

```

C:\Users\Taha's Lappy\OneDrive\Documents\oop lab 7\AI-MODEL-CARD.exe

```

-----AI MODEL CARD-----
NAME : ResNet-50
VERSION : v2.1
ALGORITHM : CNN
ACCURACY : 0.95
DEPLOYED : true
-----AI MODEL CARD-----
NAME : YOLOv8
VERSION : v8.0
ALGORITHM : CNN
ACCURACY : 0.97
DEPLOYED : true
-----AI MODEL CARD-----
NAME : SpamDetector
VERSION : v9.0
ALGORITHM : SVM
ACCURACY : 0.89
DEPLOYED : false
-----
Process exited after 0.2074 seconds with return value 0
Press any key to continue . . .

```